

ENVS-193DS-FINAL

Damian Rojas

Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see <https://quarto.org>.

Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
#package set up
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

```
chisq.test, fisher.test
```

```
library(readxl)
library(DHARMA)
```

This is DHARMA 0.5.0. For overview type '?DHARMA'. For recent changes, type `news(package = 'DHARMA')`.

Note that the default setting in `simulateResiduals` was changed to conditional simulations since version 0.5.0. For more details, see `?simulateResiduals`.

```
nests <- read.csv("nest_data_final.csv")
```

1

a)

In part 1 they used a binomial (logistical regression) test. In part 2, they used a chi-square test to check for a relationship.

b)

Part 1 : With a logistical regression we find that distance to water edge does predict the probability of American Avocet microhabitat use. Water edge proximity is an important variable American Avocet's use to determine where they live.

Part 2: We found that dependening on bird species, preference in habitat type also changes using a Chi-Square test. Habitat type preference is not consistent among varying bird species.

c)

One additional variable that could influence the probability of American Avocet microhabitat use is temperature. Temperature is a continuous variable (measured in either degrees celsius or fahrenheit) and could be indicative of microhabitat use along with water edge proximity as typically areas close to water have much more temperate climates, which could be a preference of the American Avocet.

2

a)

	bird_species	ant_species	case_control	scientific_name
1	GCSW	C. mimosae	0	Crematogaster mimosae
2	GHS	C. mimosae	1	Crematogaster mimosae
3	STAR	C. mimosae	1	Crematogaster mimosae
4	GHS	C. mimosae	0	Crematogaster mimosae
5	STAR	C. mimosae	0	Crematogaster mimosae
6	GCSW	C. mimosae	0	Crematogaster mimosae
7	STAR	C. mimosae	1	Crematogaster mimosae
8	STAR	C. nigriceps	1	Crematogaster nigriceps
9	STAR	C. mimosae	0	Crematogaster mimosae
10	STAR	C. mimosae	0	Crematogaster mimosae

b)

The 1s and 0s show whether or not a tree contains a nest (1) or is available for nesting (0)

c)

Tree height is a continuous variable. As tree height increases the probability of nest occurrence would increase because it provides more areas for birds to create nests in the area. (unable to access paper cannot cite this information, merely a guess)

Acacia Ant species is a categorical variable. The species of Acacia ant can fend off insects that harm bird offspring which increases the probability of nest occurrence. (abstract, paragraph 1).

d)

```
model_tabel <- tibble(  
  model_number = c(1,2),  
  predictors = c("bird_species + ant_species") ,  
  interaction_term = c("No", "Yes"),  
  model_description = c("Contains bird species and ant species main effects", "Contains inter  
)
```

e)

```
model0 <- lm(data = nests_clean,  
             case_control ~ bird_species + ant_species  
)  
  
model1 <- lm(data = nests_clean,  
             case_control ~ bird_species*ant_species)
```

f)

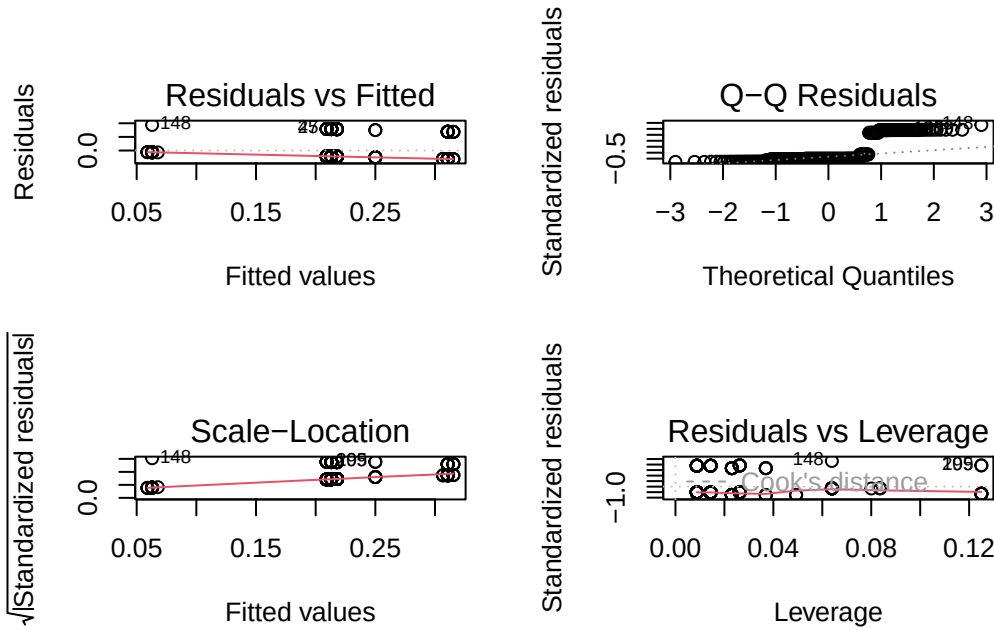
```
AIC(model0,model1)
```

	df	AIC
model0	7	301.6860
model1	11	304.5154

The best model as determined by Akaike's Information Criterion was the model predicting whether a tree was nested or not from bird species and ant species with no interaction term.

g)

```
par(mfrow = c(2,2))  
plot(model0)
```



h)

3

a)

The visualizations are different because the affective visualization tells a story about the data whereas the visualizations in homework 2 show just the raw data, no story. Within the visualizations the only pattern I can see between the both of them is that they don't really vary, the data remains similar. During week 9 workshop I hadn't created a visualization because I had submitted an extension for homework 3, so during the classmate feedback i bounced ideas off my classmates and they said that showing the route I would take was a very solid idea so I launched my visualization idea off of that interaction.

b)

Based on my test of walk time and assignment workload, I believe a linear regression would be an appropriate test because workload was measured numerically and walk time is a continuous variable, which can be predicted with the linear regression.